

NOVOBIRILYUSSY, Russian Federation

WMO: 293670

Lat: 56.9697N

Lon: 90.6831E

Elev: 593

StdP: 14.38

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -39.8	(c) -31.3	(d) -43.9	(e) 0.4	(f) -38.4	(g) -36.0	(h) 0.7	(i) -30.6	(j) 13.7	(k) 19.2	(l) 12.1	(m) 19.2	(n) 1.2	(o) 190	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.7	(c) 84.1	(d) 66.8	(e) 80.8	(f) 65.5	(g) 77.6	(h) 64.1	(i) 69.7	(j) 79.7	(k) 67.9	(l) 77.2	(m) 66.1	(n) 74.8	(o) 4.7	(p) 10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	99.1	73.9	64.5	92.9	72.4	62.7	87.1	70.7	34.2	79.7	32.7	77.5	31.3	75.0	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.5	11.1	9.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-45.6	72.4	6.1	2.9	-50.0	74.4	-53.6	76.1	-57.0	77.7	-61.5	79.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.3	0.9	8.1	20.2	36.7	48.1	60.7	64.8	59.4	46.3	32.7	13.7	5.8
	DBStd	25.01	17.26	14.30	12.55	8.47	9.84	7.75	5.70	6.69	8.14	9.81	16.53	16.79
	HDD50	7368	1523	1174	924	406	159	12	0	8	165	540	1088	1369
	HDD65	11722	1988	1594	1389	848	529	168	74	197	561	1002	1538	1834
	CDD50	1257	0	0	0	8	101	334	459	298	54	3	0	0
	CDD65	135	0	0	0	0	6	39	68	22	0	0	0	0
	CDH74	1494	0	0	0	4	90	569	630	184	17	0	0	0
	CDH80	317	0	0	0	0	14	135	132	36	0	0	0	0
(14) Wind	WSAvg	4.3	4.5	4.5	4.8	5.3	4.9	3.5	3.1	3.1	3.8	4.8	4.6	4.6
(15) Precipitation	PrecAvg	20.1	1.0	0.7	0.7	1.1	1.7	2.5	2.2	3.0	1.9	2.0	1.7	1.5
	PrecMax	25.5	2.0	1.3	1.9	1.7	3.5	4.6	4.6	6.1	4.0	3.1	2.6	2.1
	PrecMin	16.3	0.1	0.3	0.2	0.4	0.6	0.7	0.7	0.9	0.4	0.9	0.9	0.6
	PrecStd	2.6	0.4	0.2	0.4	0.4	0.8	1.0	1.0	1.3	0.9	0.6	0.4	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.6	34.9	48.1	69.1	82.5	87.5	88.1	84.8	76.7	61.9	47.2	34.6
		MCWB	30.1	30.2	39.2	51.7	60.8	68.2	69.7	68.3	60.4	48.9	42.8	31.8
	2%	DB	29.9	32.5	42.8	61.3	76.4	83.7	84.6	79.6	69.2	54.2	41.8	31.7
		MCWB	27.2	29.0	35.3	47.2	58.0	65.9	68.0	66.8	58.7	45.8	38.6	29.9
	5%	DB	26.9	30.1	40.0	55.9	71.4	80.3	80.9	75.4	64.2	49.9	38.1	28.8
		MCWB	24.8	27.4	34.0	44.9	54.9	64.1	67.6	64.6	55.2	44.0	34.4	26.9
	10%	DB	23.7	27.6	36.9	51.5	66.4	76.5	77.8	72.2	59.4	46.1	34.5	25.6
		MCWB	21.8	25.0	32.2	42.2	52.9	62.6	65.7	63.0	52.7	41.7	30.8	23.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.9	31.7	39.7	52.8	62.3	70.8	72.9	70.7	61.9	50.1	43.3	32.4
		MCDB	33.1	34.1	45.0	67.8	79.0	82.0	81.7	82.5	70.9	57.7	46.4	33.9
	2%	WB	27.4	29.5	36.6	48.8	59.3	68.1	70.7	67.7	59.2	47.4	38.8	30.0
		MCDB	29.3	31.7	41.6	59.4	71.1	79.8	80.1	76.1	68.8	53.0	41.3	31.2
	5%	WB	24.5	27.6	34.5	45.7	56.9	66.3	68.8	65.8	56.7	44.7	34.8	26.7
		MCDB	26.7	30.0	38.8	54.8	69.2	76.6	77.9	73.5	63.6	49.2	37.4	28.3
	10%	WB	21.8	24.9	32.4	42.5	54.3	63.9	67.1	63.8	53.9	42.4	31.8	23.8
		MCDB	23.8	27.5	36.7	50.4	64.9	73.7	75.6	70.5	58.5	46.6	33.8	25.4
(35) Mean Daily Temperature Range	MDBR		15.0	16.6	18.9	17.7	19.4	20.9	19.7	17.7	16.0	12.1	12.4	13.4
	5% DB	MCDBR	13.1	13.3	19.7	24.8	27.3	28.7	24.8	23.1	25.7	17.9	12.2	12.2
		MCWBR	11.8	11.4	14.0	14.0	12.7	12.7	11.6	12.3	14.2	11.2	9.9	11.3
	5% WB	MCDBR	13.0	11.6	16.4	23.4	24.2	25.1	20.7	20.4	23.6	15.7	11.5	10.8
		MCWBR	12.1	10.3	12.3	14.0	12.2	11.7	11.0	11.5	13.5	10.7	9.6	10.3
(40) Clear-Sky Solar Irradiance	taub		0.289	0.322	0.366	0.395	0.408	0.409	0.432	0.410	0.370	0.349	0.316	0.287
	taud		1.979	1.966	1.916	2.057	2.206	2.291	2.232	2.271	2.334	2.221	2.016	1.930
	Ebn at Noon		169	217	239	257	262	264	254	251	244	211	155	129
	Edh at Noon		21	32	44	45	41	38	40	36	29	25	20	16
(44) All-Sky Solar Radiation	RadAvg		224	522	938	1371	1574	1877	1714	1333	836	420	206	128
	RadStd		30	48	69	97	174	173	114	115	111	40	34	22

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page